

TUNAHAN ERBAY

Hi, I'm Tuni. I build software that is designed to last. With 6+ years of experience across frontend and backend development, I enjoy solving complex technical challenges, improving existing systems, and creating reliable solutions that deliver real value.



CONTACT

Email

tunahan@erbay.ch

Location

Obergösgen, CH

LinkedIn

www.linkedin.com/in/terbay/

GitHub

github.com/etunidev

LANGUAGES

German

Native

Swiss German

Native

Turkish

Native

English

Professional working proficiency

French

Basic written proficiency

INTERESTS

AI, Process Automation, Sports

SOFT SKILLS

Problem Solving, Communication, Teamwork,
Technical Leadership, Analytical Thinking

EXPERIENCE

► Aug 2022 – Present

Senior Software Engineer

► ELCA Informatik AG, Bern, CH

Joined ELCA as a Fullstack Software Engineer and was promoted to Senior Software Engineer in 2025. Over time, I took on additional responsibilities as Scrum Master and Lead Developer, contributing to software architecture, technical decision-making, and the successful delivery of enterprise software solutions. Currently, I am working on a new project, helping establish its frontend, backend, and DevOps foundations while driving development across the full technology stack.

- Promoted from Software Engineer to Senior Software Engineer while taking on additional responsibilities as Scrum Master and Lead Developer.
- Led agile delivery processes as Scrum Master during a major release, facilitating collaboration between Swiss and offshore development teams.
- Helped establish a new project from the ground up, including frontend, backend, DevOps, and development processes.

Spring Boot Angular SAP CC WSO2 Git Maven Jenkins Bitbucket
Jira Confluence Scrum Figma Miro IntelliJ IDEA Visual Studio Code

► Jan 2021 – Jul 2021

Web Developer

► Relaxor, Olten, CH

Built and operated an e-commerce business to gain hands-on experience in web development, digital marketing, and customer acquisition.

- Built a WooCommerce-based e-commerce platform from scratch, including frontend implementation and visual design.
- Acquired initial customers through targeted Google Ads and Meta Ads campaigns.

WooCommerce Wordpress JavaScript HTML CSS E-Commerce
Marketing Google Ads Meta Ads

► Jan 2020 – Present

Founder / Web Developer

► Webmo, Olten, CH

Developed and maintained WordPress websites and hosting solutions for private individuals and small to medium-sized businesses. Responsible for the technical implementation, customization, maintenance, and long-term support of client solutions.

- Designed and operated a Linux- and Nginx-based hosting platform for client projects.
- Developed, customized, and maintained WordPress websites and plugins for more than 20 clients.

WordPress JavaScript HTML CSS Client Support Linux Nginx

EDUCATION

► Aug 2018 – Jul 2022

Bachelor of Computer Science

► University of Basel, Basel, CH

Bachelor's degree in Computer Science with a focus on software engineering, programming, databases, algorithms, and practical software development projects.

► Aug 2014 – Jul 2018

Swiss Federal Matura

► Kantonsschule Olten, Olten, CH

Secondary education with a specialization in Physics and Applications of Mathematics.

SKILLS

Backend

Java Spring Boot Python SQL

Frontend

Angular TypeScript JavaScript HTML CSS React

DevOps

Git Docker Kubernetes k3s ArgoCD GitHub Actions Traefik CI/CD GitOps Maven Jenkins Bitbucket RHOS

Tools

IntelliJ IDEA Visual Studio Code Jira Confluence Figma Miro

Methods

Scrum SAFe BPMN Software Architecture

PROJECTS

tuni.dev — Portfolio & DevOps Platform

May 2026

Personal Project — Remote

Built and deployed a personal portfolio website with a full DevOps pipeline, covering containerization, GitOps-based continuous delivery, and cloud infrastructure on Kubernetes.

- Built a fully responsive portfolio site in React + TypeScript with Tailwind CSS, with dynamic content driven entirely by a single JSON file.
- Built a CI/CD pipeline with GitHub Actions that builds a Docker image, pushes to GHCR, and auto-updates the Kubernetes manifest on every push.

Docker Kubernetes k3s ArgoCD GitHub Actions Traefik cert-manager GitOps CI/CD React

Bachelor Thesis: Secure Scuttlebutt Kotlin Library

Jan 2022

University of Basel — Basel, CH

Developed a Kotlin-based implementation of the Secure Scuttlebutt (SSB) group chat protocol and transformed it into a reusable library, enabling secure and decentralized group communication within the Tremola ecosystem.

- Implemented the protocol's core cryptographic functionality, including key derivation, HMAC-based key generation, and secure message encryption/decryption to ensure confidential group communication.
- Designed and implemented the Slot/Unslot mechanisms and group identifier mapping, enabling secure recipient access control while preserving group privacy in public references.

Kotlin Security Encryption HMAC

Data Integration and Analysis Pipeline

Jan 2021

University of Basel — Basel, CH

Designed and implemented a data integration pipeline for collecting, transforming, and analyzing heterogeneous datasets from multiple sources.

- Analyzed 4 GB of U.S. crime and economic data drawn from multiple sources and mixed datatypes, using SQL and Pandas to integrate them into a single coherent dataset. Performed data cleaning, integration, and visualization.

SQL Python Data Engineering ETL Data Visualization Data Analysis

Linux Command-Line Integration Tool

Oct 2020

University of Basel — Basel, CH

Reimplemented core Linux shell functionality and file system operations in C, including support for remote file system integration.

- Developed a minimal Gnome-Terminal-style shell in C, reimplementing 20+ core commands and adding support for mounting remote servers over SSH so they appear as regular directories.

C Systems Programming Filesystems SSH

Decentralized Messaging System

May 2020

University of Basel — Basel, CH

Developed a secure decentralized messaging application enabling peer-to-peer communication without relying on centralized servers.

- Built an end-to-end encrypted chat application UI in Python using Tkinter, implementing decentralized peer-to-peer communication over BACnet without relying on a central server.

Python Encryption Networking Peer-to-Peer Security

Modular LED Gaming Console

Jan 2020

[University of Basel](#) — [Basel, CH](#)

Designed and built a modular LED-based gaming console from scratch, including both hardware and software components.

- Designed and built a custom multiplexed LED matrix with simulated circuitry and hand-soldered components, then programmed a plug-in game framework with joystick input in Arduino/C++.

C

C++

Circuit Design

Embedded Systems

Hardware Design

Arduino

JabRef User Interface Extension

Dec 2019

[University of Basel](#) — [Basel, CH](#)

Developed a user interface extension for the open-source project JabRef, enhancing customization and accessibility features.

- Developed and implemented a user interface extension for JabRef, a widely used open-source Java application for managing BibTeX and BibLaTeX databases. The extension provided additional customization options, including themes, color schemes, and templates, while addressing accessibility requirements and enhancing usability.

Java

Scrum

Accessibility

Multiplayer Gorillas Remake

May 2019

[University of Basel](#) — [Basel, CH](#)

Designed and developed a multiplayer remake of the classic Gorillas game, focusing on game physics, networking, and real-time synchronization.

- Implemented deterministic physics, collision handling, and a real-time peer-to-peer networking layer in Java for synchronized multiplayer.

Java

Game Physics

Networking

CERTIFICATIONS & COURSES

- **Developing Generative AI Applications on AWS** , Fast Lane (Jan 2026) — Generative AI, AWS
- **Leading SAFe - Certified SAFe 6 Agilist** **Certified** , Scaled Agile, Inc. (Mar 2025) — SAFe, Agile
- **Communication** , ELCA (Feb 2025) — Communication
- **Spring & Hibernate for Beginners** , Udemy (Nov 2024) — Spring, Hibernate, Spring Boot
- **Professional Scrum Master I** **Certified** , Scrum.org (Aug 2024) — Agile, Scrum
- **Problem Solving** , ELCA (Jun 2024) — Problem Solving
- **Kubernetes Native Development with Quarkus** , Red Hat (May 2024) — Kubernetes, Quarkus, Architecture
- **The Role of Scrum Master** , ELCA (Oct 2023) — Scrum
- **Usability/UX Introduction** , ELCA (Sep 2023) — UX, Usability
- **Angular Advanced Training** , Jonas Bandi (Jun 2023) — Angular, TypeScript
- **Quality in Software Design** , ELCA (May 2023) — Software Design, Quality
- **Code Quality** , ELCA (May 2023) — Code Quality
- **BPMN Method & Style** **Certified** , Stephan Fischli (Mar 2023) — BPMN
- **Application Security Fundamentals** , ELCA (Feb 2023) — Application Security
- **Angular - The Complete Guide** , Udemy (Sep 2022) — Angular, TypeScript, JavaScript